UNIT 4 · 9 hours

BJT: Construction and Operation



Transistors

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

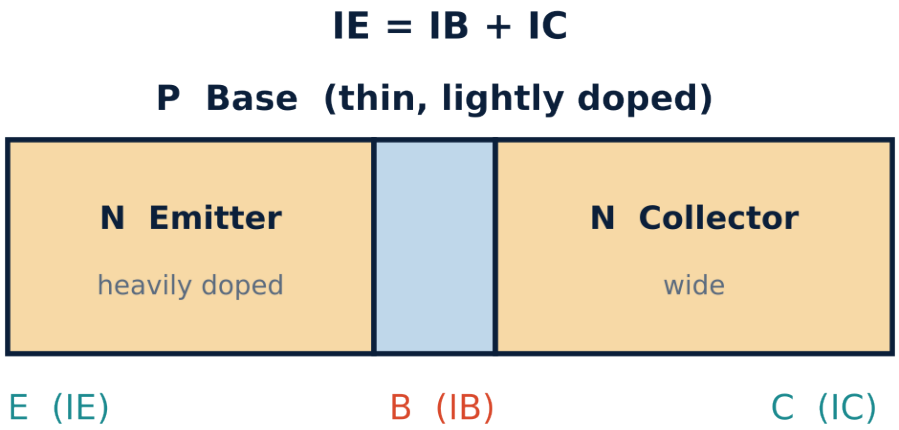
- Describe the construction of npn and pnp transistors
- Explain biasing for active-mode operation
- Relate IE, IB, IC and the gains α and β
- Distinguish CB, CE and CC configurations

Syllabus context

Construction and operation, BJT as a switch, working and characteristics of n-channel enhancement type MOSFET, common source amplifier, DC load line analysis, MOS inverter.

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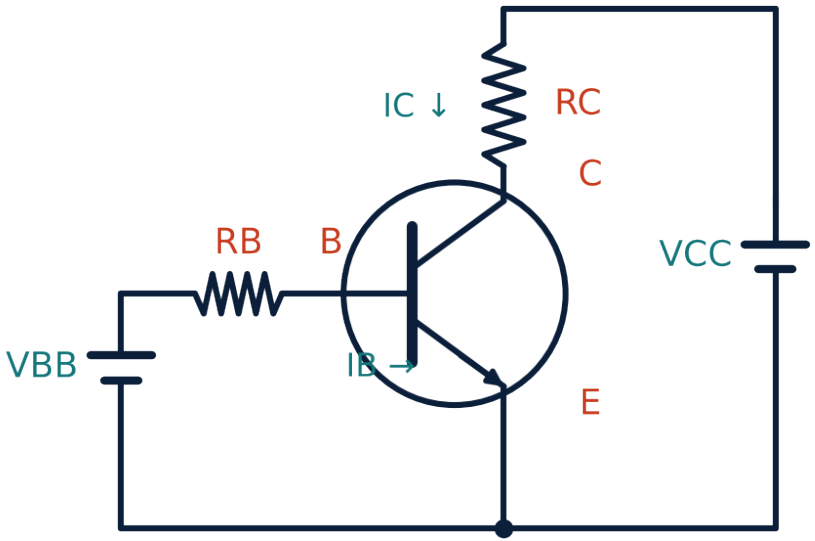
Construction



- Three regions, two PN junctions: npn or pnp.
- Emitter: heavily doped; emits carriers.
- Base: very thin, lightly doped.
- Collector: widest region; collects carriers.
- Invented by Bardeen, Shockley and Brattain (1948).

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Biasing the npn transistor



- Emitter–base junction forward biased: VBE ≈ 0.7 V (Si), 0.3 V (Ge).
- Collector–base junction reverse biased: VCC > VBB.
- RB, RC are current-limiting resistors.
- Collector voltage typically 3–20 V.

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What happens inside (npn, active mode)

1

Injection

Forward-biased emitter junction injects electrons into the base.

2

Thin base

Only a few (1–5 %) recombine with holes: this is the base current IB.

3

Collection

The rest are swept across the reverse-biased junction into the collector: IC.

4

Leakage

Reverse minority-carrier current ICBO adds slightly to IC.

IE = IB + IC (Kirchhoff's current law)

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Current gains

$$\alpha = I_C / I_E$$

0.95 – 0.99

$$\beta = I_C / I_B$$

20 – 500

$$\beta = \alpha / (1 - \alpha)$$

relation

$$I_E = (\beta + 1)\, I_B$$

emitter current

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Transistor configurations

Configuration	Common terminal	Gain	Typical use
Common base (CB)	Base	Voltage gain, no current gain	High-frequency stages
Common emitter (CE)	Emitter	Both current and voltage gain	Amplifiers, switches
Common collector (CC)	Collector	Current gain, no voltage gain	Buffers (emitter follower)

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Numerical: currents and gains

Problem A transistor has IB = 0.08 mA and IE = 9.60 mA. Find IC, α and β.

- IC = IE − IB = 9.60 − 0.08 = 9.52 mA
- α = IC/IE = 9.52/9.60 = 0.9917
- β = IC/IB = 9.52/0.08

Answer: IC = 9.52 mA, α = 0.9917, β = 119

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Key takeaways

- Emitter heavily doped, base thin, collector widest
- Active mode: EB forward, CB reverse
- $I_E = I_B + I_C$; $\alpha = I_C/I_E$; $\beta = I_C/I_B$
- CE has both current and voltage gain

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Practice questions

Q1. Why is the base region made thin and lightly doped?

Q2. IC = 5 mA, β = 100. Find IB and IE.

Q3. Find β for α = 0.98.

Q4. Compare CB, CE and CC configurations.